Three-parameter proof calculi and modular saturation

A measured alternative that does not beat the natural-variable parent

Research artifact for ProveIt

25 October 2026

Abstract

We replace a four-parameter implicational axiom by K, S, and Peirce and remove its companion bounded disequality witness using an unbounded modular inverse. A small-value decoding argument makes the inverse usable without another binomial range certificate. We prove both directions of this change, including a necessary lift of two encoded representatives to keep the saturation quotient nonnegative. A literal 521-step Hilbert derivation certifies the propositional replacement. The implementation produces a quartic in 231 nonnegative integer variables with 183 quadratic residuals; it loses to the retained 229-variable natural parent. It is an experimental source-calculus alternative, not the preferred equation of the surrounding package and not a new end-to-end formalization.

1 Scope, parent, and exact outcome

The parent is the natural-variable constructor in `ConZFC_Residual_Bases_Denominator_Engineering_2026-10-25.zip`, retained in ProveIt at commit `3ecd6a6cf53d47519228328567f45fb7ee9781d9`. The archive and all required compiler files are identified by exact hashes in `PARENT_PIN.json`. The parent uses a numerical proof calculus calibrated to ZFC, descending factorials, reciprocal CRT moduli, binomial digit certificates, a shared Pell frame, and one semantic index anchor. We retain that arithmetic architecture and change the local source only.

All coordinates here range over $\mathbb{N} = \{0, 1, \ldots\}$. The delivered equation is

$$P_{231}(z_0, \ldots, z_{230}) = \sum_{j=0}^{182} R_j(z_0, \ldots, z_{230})^2 = 0,$$

where every coefficient of every quadratic R_j is in `equation.json`. The fully collected quartic is in `expanded_polynomial.json`. No exponentiation, binomial, factorial, bounded quantifier, or proof predicate is a primitive of this polynomial. The following metrics are recomputed from its literal coefficients.

Natural endpoint	Variables	Rows	Degree	Monomials	Height
Retained projected parent	229	179	4	6,596	83,879,378
This experiment	231	183	4	6,369	83,899,166

The parent also has a balanced row-basis endpoint with 5,607 monomials and height 65,892,965. Thus even the lower monomial count in the displayed comparison is not a sparsity record. The main construction elsewhere in this package uses unrestricted integers; its coordinate count is a different comparison. The present experiment supplies a source-level theorem and a fully measured unsuccessful compression attempt.

2 A basis with at most three parameters

Write the numerical implication constructor as

$$I(a,b) = 2(a+b)(a+b+1) + 4b,$$

and, temporarily, write $a \to b$ for this constructor. The source calculus has the four-parameter axiom

$$((c \to d) \to a) \to ((a \to c) \to (b \to c)). \tag{1}$$

Replace it by the three schemata

$$\begin{aligned} \mathrm{K}: &\quad a \to (b \to a),\\ \mathrm{S}: &\quad (a \to (b \to c)) \to ((a \to b) \to (a \to c)),\\ \mathrm{P}: &\quad ((a \to b) \to a) \to a. \end{aligned}$$

Retain all other numerical axiom families, the guarded bundle, modus ponens, and generalization. No new bound is imposed on the numerical instantiations.

Proposition 1. *The original and replacement numerical calculi have the same derivable codes.*

Proof. The retained calibration source `KCode222.lean` contains `c222_k`, `c222_s`, and `c222_classical`, whose Peirce component is supplied by `Lukasiewicz.peirce`. Thus each new schema is derivable in the original calculus.

For the converse, K and S support the ordinary deduction transformation. If an axiom or an undischarged assumption proves D, K derives $A \to D$. The discharged assumption A is treated by the standard K/S proof of $A \to A$. A modus-ponens step is discharged with S. This is a finite syntactic transformation, valid under every interpretation of the binary constructor.

Put $f = (c \to d) \to a$ and $g = a \to c$. Under assumptions $f, g, c \to d$, two applications of modus ponens derive c. Discharging $c \to d$ gives $(c \to d) \to c$. Peirce with parameters c, d then gives c under f, g. Add the unused assumption b and discharge b, g, f, in that order. The result is precisely (1).

The supplied certificate expands all deduction transformations into 521 literal lines: 132 K instances, 128 S instances, one Peirce instance, and 260 modus-ponens steps. The independent checker accepts only these axiom instances and exact earlier-line references. It accepts no remaining assumptions or deduction-theorem rule. Replacing every use of an axiom by its finite derivation now gives both inclusions of derivable codes. □

The source's bundle requires $c \neq a$ and $c \neq b$. Its parameter c also occurs as the formula matrix, so simply choosing $c = a + b + 1$ is not an equivalent axiom restriction. We keep c unrestricted and enforce its two disequalities by the following separate device.

3 The modular saturation lemma

Let $\boldsymbol{z}$ denote the bounded local fields. Let $U(\boldsymbol{z})$ be a product of squared residuals, one for each *non-bundle* branch. Let $V(\boldsymbol{z})$ be the squared residual for the unguarded bundle transition, and put

$$D(\boldsymbol{z}) = (c-a)^2(c-b)^2, \qquad L(\boldsymbol{z}) = U(\boldsymbol{z})V(\boldsymbol{z}) + \sum_{\nu} G_\nu(\boldsymbol{z})^2.$$

Here the G_ν are the two beta-sequence residuals. Consequently U, V, L are nonnegative at every natural tuple. The desired local condition is

$$G_\nu = 0 \text{ for every } \nu, \qquad U = 0 \quad \text{or} \quad (V = 0 \text{ and } D \neq 0). \tag{2}$$

Let $\boldsymbol{A}$ be an encoded tuple, with a common modulus N. Add just the two global equalities

$$L(\boldsymbol{A}) = Nq_L, \qquad U(\boldsymbol{A})D(\boldsymbol{A})e = U(\boldsymbol{A}) + Nq_s. \tag{3}$$

The new e is an *unbounded* natural coordinate. It does not receive a range certificate or become a Pell sequence index.

Lemma 2 (Soundness)**.** *Suppose $m \mid N$, $m \geq 2$, and $\boldsymbol{A} \equiv \boldsymbol{z} \pmod{m}$. If* (3) *holds and*

$$0 \leq L(\boldsymbol{z}) < m, \qquad 0 \leq U(\boldsymbol{z}) < m,$$

then (2) *holds, regardless of the size of e.*

Proof. The first equality implies $L(\boldsymbol{z}) \equiv 0 \pmod{m}$; its bound therefore gives $L(\boldsymbol{z}) = 0$. Nonnegativity gives all $G_\nu = 0$ and $UV = 0$. If $D(\boldsymbol{z}) = 0$, reduce the second equality modulo m. It gives $U(\boldsymbol{z}) \equiv 0 \pmod{m}$, and the separate bound gives $U(\boldsymbol{z}) = 0$. Otherwise $D \neq 0$, and $UV = 0$ gives the stated alternative. □

The separate upper bound on U is essential: a bound on UV does not bound U when $V = 0$. For example, $m = 3$, $U = 3$, $V = D = 0$ satisfies both zero congruences with an invalid guard. The construction's majorant explicitly includes U, in addition to L, rather than assuming this missing bound.

Lemma 3 (CRT completeness)**.** *Let $m_1, \ldots, m_t$ be pairwise coprime, $K \geq 1$, and $N = (\prod_i m_i)^K$. Suppose each local tuple $\boldsymbol{z}_i$ satisfies* (2)*. Assume $\gcd(D(\boldsymbol{z}_i), m_i) = 1$ whenever $U(\boldsymbol{z}_i) \neq 0$. Choose an encoded tuple satisfying $\boldsymbol{A} \equiv \boldsymbol{z}_i \pmod{m_i^{2K}}$ for every i. If $D(\boldsymbol{A}) \geq 1$, then positive e and natural q_L, q_s exist satisfying* (3)*.*

Proof. For every row with $U(\boldsymbol{z}_i) \neq 0$, choose $e_i \equiv D(\boldsymbol{z}_i)^{-1} \pmod{m_i^{2K}}$. For every other row choose $e_i = 0$. CRT supplies an e in these residue classes, and adding a sufficiently large multiple of N^2 makes $e \geq 1$. For every i, either $U(\boldsymbol{z}_i) = 0$ or $D(\boldsymbol{z}_i)e_i = 1$ modulo m_i^{2K}. Therefore N^2 divides $U(\boldsymbol{A})(D(\boldsymbol{A})e - 1)$. The same polynomial-congruence argument gives $N^2 \mid L(\boldsymbol{A})$. Since $L, U \geq 0$, $D(\boldsymbol{A}) \geq 1$, and $e \geq 1$, both quotients in (3) are nonnegative integers. □

Nonzero guards need not be units for arbitrary moduli. For example, $D = 3$ has no inverse modulo 3. The hypothesis is supplied here by the specific reciprocal moduli, not by an invalid general converse to Lemma 2.

4 Application to the reciprocal CRT compiler

After the inherited factorial decoding, let

$$t \geq 12, \quad F = t!, \quad m_i = F^2/i - 1 \ (1 \leq i \leq t), \quad B = \prod_{i=1}^{t} m_i = \binom{F^2 - 1}{t}, \quad N = B^x.$$

Every i divides F, so $m_i \equiv -1 \pmod{F}$. Every prime dividing an m_i is therefore greater than t. If a prime divided both m_i and m_j, it would divide $im_i - jm_j = j - i$, which is impossible for $i \neq j$. Thus the moduli are pairwise coprime. Since $0 \leq a, b, c \leq t$ in each decoded row, both nonzero differences $c - a, c - b$ have absolute value at most t. Their squared product is consequently a unit modulo the row modulus. This proves the unit hypothesis of Lemma 3.

The bounded local columns are now exactly

$$a, b, c, S, S', u, v.$$

The original eighth column d was used by the four-parameter axiom and the equality witnessing the bundle's disequality guard. Proposition 1 and Lemmas 2–3 remove both uses. The global index representative and derived-label sum retain their own range requests. Together with the exact factorial binomial this gives ten modular digit requests, one fewer than in the parent. There are six exact power graphs, twelve independent Pell seeds, and frame offset fourteen.

For soundness, choose a prime $p \mid m_i$ and put $\mu = v_p(m_i)$. The retained one-direction prime-power range lemma decodes all local columns modulo $p^{x\mu}$. The two exact global equations (3) are divisible by that modulus. The new majorant includes L, U, D and the label sum. Its computed exponent is 6,893,371, rounded upward to $E = 2^{23}$, exactly the parent's existing budget. With $R = 2 + t + x + y + F$, the inherited endpoints give

$$R < 2x, \quad x > 2^{32}, \quad L, U < R^E < (2x)^E < 2^x \le p^{x\mu}.$$

Lemma 2 applies. The unchanged beta compatibility and multiplicative Horn-flow argument now recover a refutation in the new calculus, and Proposition 1 converts it into an original one.

For completeness, convert an original refutation to the new basis and pad the horizon beyond all its local parameters and labels. The inherited beta construction supplies x, y. Choose each positive independent encoded representative initially in $(0, N^2]$, with the prescribed local residues modulo m_i^{2x}. Then replace

$$A_a \longmapsto A_a + 2N^2, \qquad A_b \longmapsto A_b + N^2, \tag{4}$$

leaving A_c in $(0, N^2]$. Both lifted representatives are strictly greater than A_c, so the global D is positive. All row congruences are preserved. Lemma 3, with $K = x$, supplies e, q_L, q_s.

The two lifts are not optional bookkeeping. In a proof that never needs a bundle branch, canonical global representatives can have $A_b = A_c$ while global $U > 0$. Then the requested natural quotient would satisfy $0 = U + Nq_s$, which is impossible. Lifting only A_a, as in the parent, would not fix this case. The metadata and proof both use (4).

The parent's radix is self-sized:

$$q = (NH)^x, \qquad H = 1 + \text{sum of the independent representatives.}$$

There is no aggregate upper-bound certificate restricting the lifts. After (4), every independent representative is at most $3N^2$. The semantic label anchor J contains the term A_a and its quadratic syntax growth; in particular $J \ge 2A_a^2 > 8N^4 > 6N^2$. It still exceeds twice every nonanchor index. The retained cofinal-anchor construction can therefore choose its Pell witnesses. The radix is chosen after the lifted representatives, so its coefficient bounds adapt to them. The unbounded e is absent from the range and Pell-index lists. No circular enlargement of the radix or horizon is introduced.

The stronger precision m_i^{2x} supplies each binomial range divisibility to precision m_i^x: one vanishing factor has the stronger valuation, whereas the denominator factorial loses at most $k - 1 \le x - 1$ for a range length $k \le x$. This is the retained one-direction argument; no general converse of the binomial range lemma is used.

5 Polynomial export and calibration boundary

The raw adapter emits 337 natural variables and 289 quadratic residuals. The inherited nonnegative affine and certified quadratic contractions eliminate 106 variables and produce the delivered 231-variable quartic. Two restorations have negative coefficients but carry exact rational sum-of-squares plus nonnegative-monomial certificates. The inherited independent checker replays those restorations and independently expands the target polynomial. The

supplied standalone checker independently checks the literal target's coefficients and the 521-line basis derivation, but does not reconstruct the inherited arithmetic interpretation.

The ordinary proof dependency is explicit. The parent supplies the descending-factorial bootstrap, binomial digit graph, common Pell frame, semantic index anchor, and original numerical-calculus calibration. The new claims proved here are the basis replacement, the modular saturation argument, and its positive-quotient/cofinal completeness adjustment. Together these give, within ZFC and relative to those retained arithmetic proofs,

$$(\exists \boldsymbol{z} \in \mathbb{N}^{231})\ P_{231}(\boldsymbol{z}) = 0 \quad \Longleftrightarrow \quad \text{there exists a finite syntactic ZFC refutation.}$$

Negating gives the requested form with Con(ZFC). This is an equivalence, not a proof of consistency. We do not claim that the new equation's entire interpretation has been checked by Lean.

The parent retains named calibration theorems `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. Its calibration files are pinned at repository commit `08200eb10fde2db408c69dffd840d6920a569b1f`; that older calibration pin is distinct from the current intake commit and is not silently updated.

6 What was learned

The inverse can safely remain unbounded because soundness reduces the guard failure to a separately bounded nonnegative factor. This is the useful general mechanism. One range column and one Pell seed disappear, but K/S/Peirce introduce additional source branches and arithmetic gates, and the saturation row also requires gates and a quotient. With this compiler the net result is two additional natural variables and four additional rows. Fewer source parameters do not imply a smaller literal Diophantine equation.

No literature-priority claim is made for modular saturation. We have not asserted an optimal implicational basis, an optimal inverse encoding, or a new variable record. The retained artifacts make the architectural change auditable and leave a precise target for later work: reduce the polynomial cost of a three-parameter classical basis, or share the inverse with an already cofinal witness using an independently proved joint existence argument. Such sharing is not used in the delivered equation.

References

[1] V. Reshetnikov, *ProveIt*, retained incoming archive `ConZFC_Residual_Bases_Denominator_Engineering_2026-10-25.zip`. `https://github.com/VladimirReshetnikov/ProveIt/tree/3ecd6a6cf53d47519228328567f45fb7ee9781d9/docs/incoming`.

[2] ProveIt calibration sources, `KCode222.lean`, `KEndpoint.lean`, and `ConstructibleChoice.lean`, retained at commit `08200eb10fde2db408c69dffd840d6920a569b1f`. `https://github.com/VladimirReshetnikov/ProveIt/tree/08200eb10fde2db408c69dffd840d6920a569b1f/Computability/BusyBeaver/Lean/BusyBeaver/ZFC`.